WISTLS Sampling Drill Design and Development

Testing

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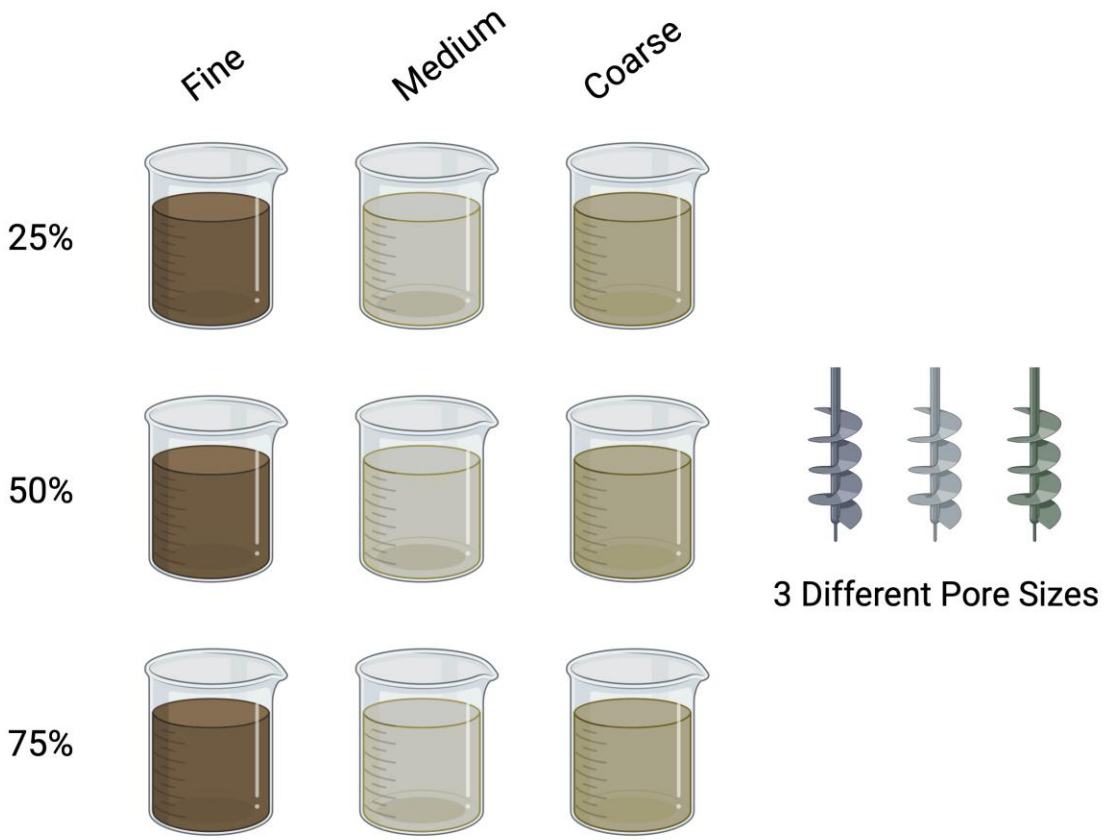
Partnered with:

Jet Propulsion Laboratory
California Institute of Technology

Kalind Carpenter - JPL
Lance Christensen - JPL
Kevin Schwarm - JPL

Cal Poly Pomona

Overarching Experiment & Design



3 Different Pore Sizes

Independent Variables

- **Particle Size:** fine, medium, coarse
- **Moisture Content:** 25%, 50%, 75%
- **Pore Size Variation:** fine, medium, coarse

Objectives

- Characterize how porosity influences water extraction efficiency and clogging behavior
- Determine optimal average moisture extraction time
- Assess auger uptake and displacement efficiency across nine soil conditions

Goal: Understand how soil composition and moisture affect extraction performance

Testing Procedure

1. Collect Soil Sample

1. Insert auger into dirt while rotating
2. Fill auger completely

2. Set Up Condensing System

1. Attach condenser tube
2. Place receptacle on scale and record initial weight

3. Heat Sample

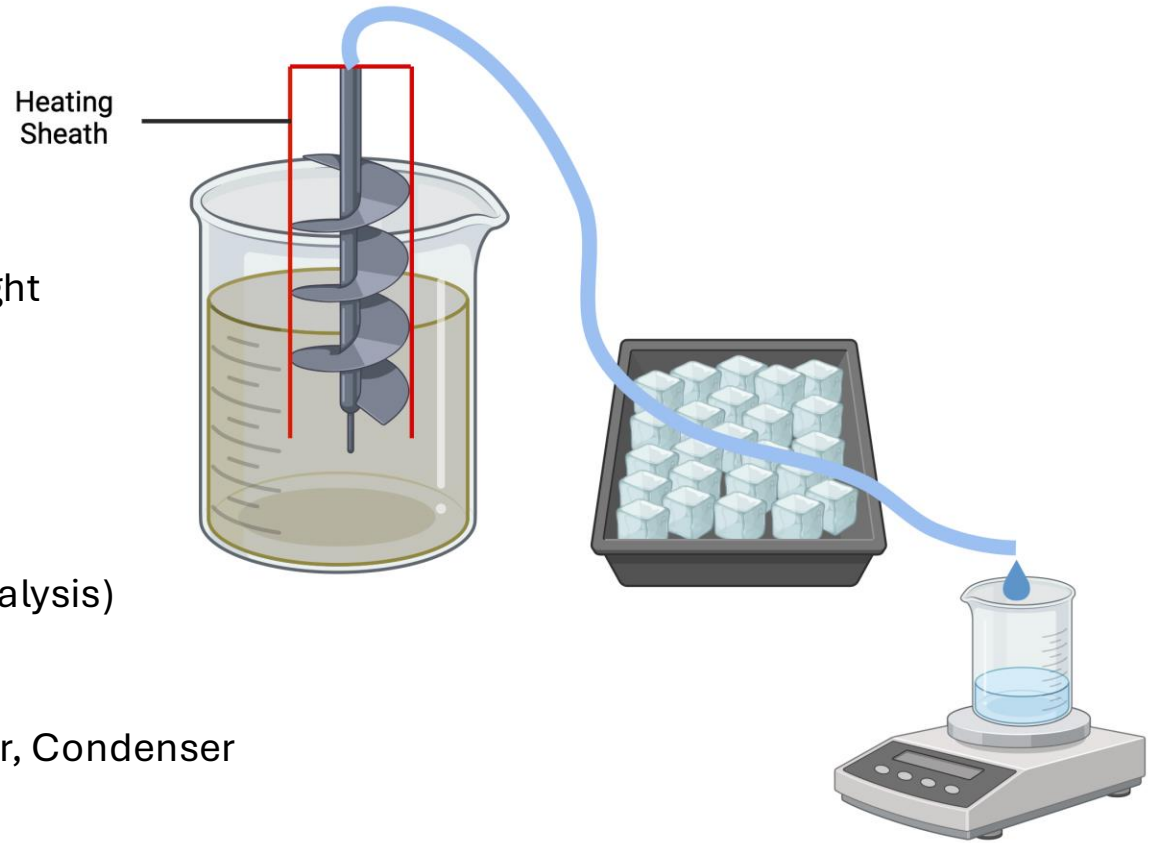
1. Heat for **X minutes** (from calibration test)
2. Measure water weight at defined time intervals
3. Identify time when extraction levels off (chart analysis)

4. Equipment Needed

1x Bucket, Auger, Drill bit, Porous tube, Hydrometer, Condenser

5. Run Experiment for X Minutes

1. Collect total water extracted
2. Use results to understand soil moisture + hardness behavior



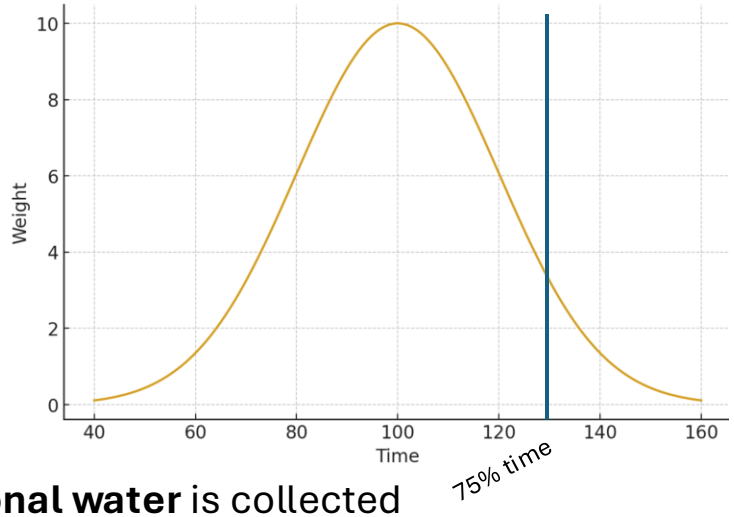
Experimental Setup
(Heating + Condenser System)

Selecting the Time Interval

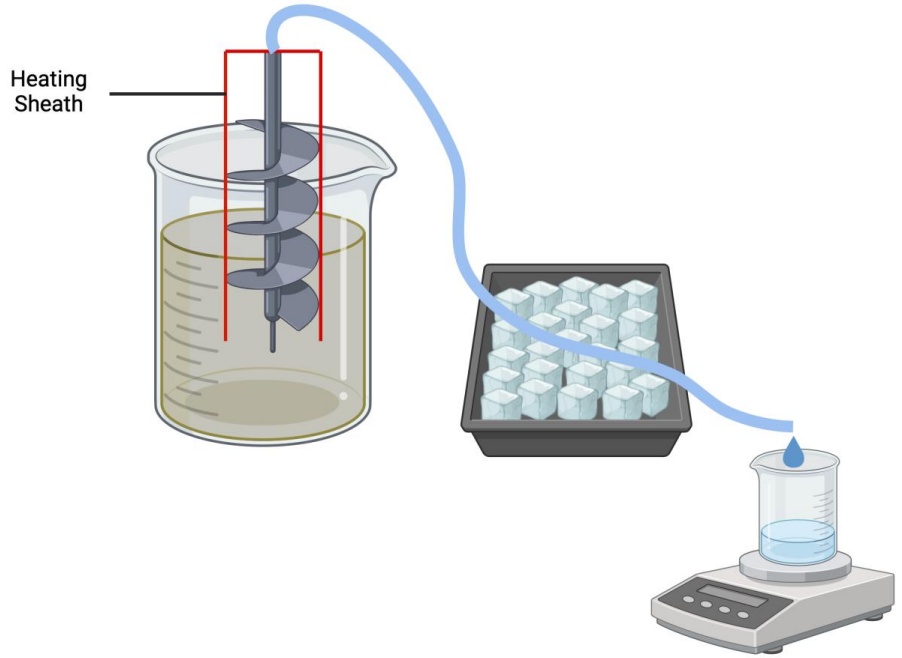
Calibration Test

- Place auger in controlled 75% humidity bucket
- Attach condenser tube to porous tube
- Collect water in a weighed glass container
- Heat and record water at increasing time intervals

Time	Weight
5 sec	
10 sec	
2min	
Etc.	



- Continue until **0% additional water** is collected
- **Outcome:** Determine total extraction time
- **Use:** Select **75% extraction time** as standard test duration



Experimental Setup
(Heating + Condenser System)

Humidity Calibration Test

Prepare container

Weigh empty container.

Add sand

Add sand and record total weight.

Ensure enough sand is added to fully submerge the auger.

Calculate required water

Use sand weight to calculate how much water is needed to reach the target humidity level.

Add water & mix

Thoroughly mix until moisture is evenly distributed.

Measure humidity

Use hydrometer to confirm the final humidity level.

Repeat for reliability

Perform **3 replicate tests** to check whether 25% humidity can be produced accurately and consistently.

Outcome: A reproducible sand humidity preparation method





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Thank You!